Removable energy pinches in the recoil graph and a direct two-point calibration

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Abstract

The internal Barnes integral of the pure-recoil tetrahedron-complement graph has energy-transfer pinch poles, including fourth-order poles on its symmetric slice. We regularize all twelve left–right Gamma pairs and prove that the resulting function is entire in the energy transfer. The complete vertex product cancels the pinch factors exactly after Schwarz pairing. A separate fixed-mass derivation of the direct C_0 two-point factor retains energy poles. These are statements about specified recoil-limit functions, not a physical four-point insertion prescription or a finite-G result.

1 Object and pinch divisors

We use the CLP graph [1], at radius one, taking $\kappa \to \infty$ at fixed real masses first. Complex masses below mean continuation of that limiting function. No interchange with a finite-κ growing-mass limit is asserted. Set

$$\boldsymbol{\mu} = (m_1, \nu, m_2, m_3, \nu, m_4), \qquad m_{1,3} = M + \omega/2 \pm u, \qquad m_{2,4} = M - \omega/2 \pm v. \tag{1}$$

The vertex sets are $123, 156, 246, 345$, and the quadrilateral sets are $1245, 1346, 2356$. Their half-sums are a_j and r_l; $\sum_j a_j = \sum_l r_l$. The previously derived recoil integral is

$$\mathfrak{B} = \frac{1}{2\pi i} \int_{d-i\infty}^{d+i\infty} e^{-i\pi z} \frac{\prod_{j=1}^{4} \Gamma(z + i a_j) \prod_{l=1}^{3} \Gamma(\frac{1}{2} - z - i r_l)}{\Gamma(z + \frac{1}{2})} \, \mathrm{d}z, \qquad 0 < d < \tfrac{1}{2}. \tag{2}$$

This line applies to the real starting data. Thereafter the contour must separate the transported pole families. A left pole $-ia_j - n$ and a right pole $\frac{1}{2} - ir_l + m$ meet when

$$p_{jl} := \tfrac{1}{2} + i(a_j - r_l) = -N, \qquad N = n + m \geq 0. \tag{3}$$

For equal probes,

$$a_j = M + \nu/2 + (\pm u \pm v)/2, \quad r_s = M + \nu + s\omega/2 \ (s = \pm 1), \quad r_0 = 2M.$$

Writing $w_j = M + \nu - a_j$, the moving divisors are

$$p_{js} = \tfrac{1}{2} - i w_j - i s\omega/2, \qquad \omega_{jsN} = -s[\, 2w_j + i(2N+1)\,]. \tag{4}$$

The signs are correlated. For $w_j > 0$, the positive-real-part upper-half-plane poles belong to $s = -1$; the opposite chirality has the conjugate families. The r_0 divisors are fixed and nonsingular for real parameters. If q of the a_j's coincide, the corresponding pole order is at most q, and is q when its leading coefficient does not vanish.

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2 An entire regularized function

Define

$$G = \prod_{j,l} \Gamma(p_{jl}), \qquad H = \mathfrak{B}/G. \tag{5}$$

For distinct real a_j, closing (2) to the left gives

$$H = \sum_j A_j \prod_{\substack{k\neq j \\ l=1,2,3}} \frac{1}{\Gamma(p_{kl})}\, {}_4F_3\left(\begin{matrix}\frac{1}{2}+ia_j, p_{j1}, p_{j2}, p_{j3} \\ (1+i(a_j-a_k))_{k\neq j}\end{matrix}; 1\right), \tag{6}$$

$$A_j = \frac{e^{-\pi a_j}\prod_{k\neq j}\Gamma(i(a_k-a_j))}{\Gamma(\frac{1}{2}-ia_j)}. \tag{7}$$

Parenthesized subscript lists in the hypergeometric denominator denote the listed parameters, not shifted factorials. The excess (sum of lower parameters minus sum of upper parameters) is exactly one in every term.

Proposition 1. *For fixed real M, ν, u, v, the continuation of H from real ω is entire in ω, including coincident a_j's.*

Proof. For distinct real a_j, the lower parameters in (6) have real part one and are independent of ω. The two varying upper parameters have constant sum. Gamma-ratio estimates give a uniform $O(n^{-2})$ bound for the series terms on every compact ω set. Finite terms are polynomials in the upper parameters, including their terminating values. Thus the series is entire in ω, as are the reciprocal Gamma factors. This uses ordinary hypergeometric convergence and Gamma properties [4, 3].

For coincident a_j, perturb real u, v in a generic direction. Near zero perturbation the expression has only finite-order Laurent singularities: the lower hypergeometric parameters remain near one, while only the Gamma differences can have poles. Each Laurent coefficient is entire in ω. Near real ω, the original vertical integral and G^{-1} are holomorphic in the perturbation, with a common exponential tail bound. Hence all negative Laurent coefficients vanish there. The identity theorem makes them vanish for every ω. The coincident limit is therefore removable. □

For computation near coincident a_j, right residues provide a better representation:

$$H = \sum_l D_l \prod_{\substack{j=1,2,3,4 \\ h\neq l}} \frac{1}{\Gamma(p_{jh})}\, {}_4F_3\left(\begin{matrix}p_{1l}, p_{2l}, p_{3l}, p_{4l} \\ 1-ir_l, (1+i(r_h-r_l))_{h\neq l}\end{matrix}; 1\right), \tag{8}$$

$$D_l = \frac{-ie^{-\pi r_l}\prod_{h\neq l}\Gamma(i(r_l-r_h))}{\Gamma(1-ir_l)}. \tag{9}$$

This formula is used away from collisions of the right families; one takes the full continued sum at any apparent singularity. The $-i$ includes $e^{-i\pi/2}$; the clockwise closing sign cancels the sign of the right Gamma residue. Reflection of the three right Gamma factors leaves a z^{-2} Gamma ratio. On closing lines chosen away from poles the sine factors give exponential decay in both imaginary directions. The remote line therefore vanishes. The excess is again one. At a pinch, one or more upper parameters are nonpositive integers, making the relevant term terminate.

For a generic simple pole, $\partial_\omega p_{js} = -is/2$. Its residue is explicitly

$$\operatorname*{Res}_{\omega_{jsN}} \mathfrak{B} = \frac{2is(-1)^N}{N!} A_j \prod_{h\neq s}\Gamma(p_{jh})\, {}_4F_3\left(\begin{matrix}\frac{1}{2}+ia_j, (p_{jl})_{l=1,2,3} \\ (1+i(a_j-a_k))_{k\neq j}\end{matrix}; 1\right)_{p_{js}=-N}. \tag{10}$$

There is no division by a vanishing leading value in our general entireness argument.

3 Exact cancellation in the complete graph

The continued product of the four recoil vertices is built from

$$T(\mu_1,\mu_2,\mu_3)=\pi^{-3/2}\sqrt{\prod_{e=1}^{3}\frac{\mu_e}{\sinh\pi\mu_e}}\prod_{e=1}^{3}\cosh\frac{\pi(S-2\mu_e)}{2},\qquad S=\sum_e\mu_e. \tag{11}$$

The twelve quantities r_l-a_j are precisely the twelve half triangle deficits $(S_v-2\mu_e)/2$. Each edge occurs twice, so the square roots disappear from the full vertex product. With

$$L(\boldsymbol{\mu})=\prod_{e=1}^{6}\frac{\mu_e}{\sinh\pi\mu_e}, \tag{12}$$

we obtain

$$\prod_v T_v=\pi^{-6}L\prod_{j,l}\sin(\pi p_{jl}). \tag{13}$$

Schwarz continuation is $f^\sharp(z)=\overline{f(\bar z)}$; at fixed real transverse data $p^\sharp=1-p$. The reflection formula then gives the exact identity

$$\boxed{\mathcal{Z}_{\rm rec}:=(\prod_v T_v)\mathfrak{B}\mathfrak{B}^\sharp=\pi^6 L\,HH^\sharp.} \tag{14}$$

Indeed $GG^\sharp=\pi^{12}/\prod\sin(\pi p_{jl})$. All the internal pinch Gamma factors cancel, with their full multiplicities. A normalized graph is obtained by dividing both sides by the same nonzero reference value. This does not introduce a physical disconnected normalization.

The regular value at a former pinch need not vanish. Cancellation of poles also does not remove either large-mass Bessel branch found in the earlier paired-graph analysis. It reorganizes the same function, rather than changing it.

For fixed positive ν, the only remaining *possible* energy poles are those of the four observer-edge factors:

$$\omega=-2(M\pm u)+2in,\qquad \omega=2(M\pm v)-2in,\qquad n\in\mathbb{Z}\setminus\{0\}. \tag{15}$$

The ratio $\mu/\sinh\pi\mu$ is regular at $\mu=0$. Some of the displayed poles may be removed by zeros of $HH^\sharp$; we do not classify them here. In particular the graph is holomorphic throughout

$$|\operatorname{Re}\omega|<2[\,M-\max(|u|,|v|)\,],\qquad M>\max(|u|,|v|). \tag{16}$$

If $M>2\nu$ and $|u|+|v|<\nu$, all internal pinch columns lie in this strip. Holomorphy alone does not equate infinite integration cycles: their ends and the external weights still require control.

4 A fixed-mass direct-C_0 calibration

CLP (3.35) uses a direct $C_0C_0^\sharp$ factor, observer weights and an insertion normalization $\mathcal{N}$ [1]. Its fixed-angle heavy-heavy-light prescription is not uniform in the fixed-mass recoil limit. We instead derive the mass-dependent ratio directly from the same small-b limit.

For a triple of masses write

$$\bar z_0=(1-iS)/2,\quad \bar z_a=(1+i(S-2\mu_a))/2,\quad z_a=1-\bar z_a,\quad L_\kappa=-i\kappa.$$

The four double-sine arguments in V^2 are $b\bar z_0-1/b$ and $b\bar z_a+1/b$. Using the exact shift relation and its fixed-parameter limit,

$$S_b(bz+1/b)=2\sin(\pi z)S_b(bz),\qquad [2\sin\pi(z-L_\kappa)]^{-1}\sim -ie^{-\pi\kappa+i\pi z},$$

gives the mass-dependent recoil factor

$$v(\boldsymbol{\mu}) = e^{\pi S/2}\Gamma(\bar{z}_0)\prod_a \Gamma(z_a)^{-1}. \tag{17}$$

Here $\bar{z}_0 + \sum_a \bar{z}_a = 2$, so the other small-$b$ powers cancel in reference-mass ratios. The previously established $U = V^2/C_0$ factor tends, in the same ratio sense, to

$$F(\boldsymbol{\mu}) = \frac{\prod_a \Gamma(1 - i\mu_a)}{\prod_a \Gamma(z_a)^2}.$$

Consequently

$$c(\boldsymbol{\mu}) = e^{\pi S/2}\frac{\Gamma(\frac{1}{2} - \frac{iS}{2})\prod_a \Gamma(\frac{1}{2} - \frac{i}{2}(S - 2\mu_a))}{\prod_a \Gamma(1 - i\mu_a)}, \tag{18}$$

$$D(\boldsymbol{\mu}) := cc^\sharp = \frac{\pi e^{\pi S}\prod_a [\sinh(\pi\mu_a)/\mu_a]}{\cosh(\pi S/2)\prod_a \cosh(\pi(S - 2\mu_a)/2)}. \tag{19}$$

Only reference ratios of v, c, D are fixed by the limiting derivation; common mass-independent factors have been removed. This is not an absolute normalization of physical insertions.

For $\mu_{1,2} = M \pm \delta/2$, $\mu_3 = \nu$, $M > 2\nu > 0$, the direct factor D has simple poles at $\delta = \pm\nu + i(2N + 1)$, $N \in \mathbb{Z}$. At $\delta_0 = \nu + i(2N + 1)$ its nonzero residue is

$$\operatorname*{Res}_{\delta_0} D = -\frac{2e^{\pi(2M+\nu)}\sinh(\pi\mu_1)\sinh(\pi\mu_2)}{\mu_1\mu_2\nu\cosh(\pi(2M + \nu)/2)\cosh(\pi(2M - \nu)/2)}\bigg|_{\delta=\delta_0}. \tag{20}$$

Here δ is a two-point mass difference and must be mapped explicitly before comparison with a four-point global transfer.

This contrast supplies a calibration condition, not a unique sewing rule. The complex continuation of $\mathcal{N}$, the physical inner product, the spectral measure and disconnected factorization remain to be specified. One cannot restore desired poles by arbitrarily multiplying a graph by D or its square root. The mixed representation and contour issues of CSTY [2] therefore remain relevant.

5 Numerical evidence and continuation

The saved driver performs 74 checks at 40 decimal digits, with selected 60-digit comparisons. It compares the two residue representations, the unregularized four-term sum, an original vertical integral, and the complete vertex identity. Three examples have left-sum condition numbers up to 3.68×10^4, while the corresponding right sums have condition numbers at most 1.0675.

For $M = 3, \nu = 1$, the transverse pairs $(1/8, 1/4), (0, 1/4), (0, 0)$ give simple, double and quadruple pinches. Both $N = 0, 1$ are tested. Approaches $\omega = \omega_0 + \epsilon$, with $\epsilon = 10^{-3}, 10^{-5}, 10^{-7}$, verify the predicted ϵ^{-q} behavior of $\mathfrak{B}$ and a finite full graph. The largest final relative approach errors are approximately 8.12×10^{-7} for the pole coefficient and 5.31×10^{-7} for the full graph. These are finite-ϵ effects, not quadrature error bars. For example, at the symmetric fourth-order pinch $\omega_0 = 1 + i$,

$$\mathcal{Z}_{\mathrm{rec}}(\omega_0) = -4.38134684304 \times 10^{-10} + 1.73872354359 \times 10^{-10} i.$$

The exact-pinch value is evaluated using the terminating regularized representation, not inferred solely by extrapolation.

Two mass samples also test the exact double-sine V^2 factor at $\kappa = 40, 160, 640$ against v. This tests a constituent of the C_0 derivation, not the complete finite-κ C_0 or four-point graph. No interval certificate or finite-G dynamical claim is made.

The next task is to calibrate observer weights and disconnected limits against (19), then seek an actual sewing/inner-product prescription. Identification with a known Racah function requires a parameter, measure and normalization dictionary, not just a similar Barnes integral. Internal pinch locations alone no longer select a physical external cycle; growth at infinity and physical insertions must do that work.

Reproducible records. Chinese derivation: notes/stage4-recoil-poles.md. Implementation: code/recoil_poles.py and code/validate_recoil_poles.py. Raw values, precision, condition numbers and input hashes: results/stage4/recoil-poles/diagnostics.json.

References

[1] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, especially (3.17), (3.35)–(3.37), (3.43), (3.46)–(3.47) and Appendix A.

[2] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1.

[3] NIST Digital Library of Mathematical Functions, §5.2 and §5.5.

[4] NIST Digital Library of Mathematical Functions, §16.2 and §16.5.